



# Introduction to Java

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Java is a widely used programming language for building web, desktop, mobile, and enterprise applications. It is known for its simple syntax, portability, and powerful features that make development easier and more secure.

- Supports object-oriented programming concepts like classes, objects, inheritance, and polymorphism
- Platform independent because of JVM support
- Secure and robust with strong memory management

## Features of Java

Java offers features like object-oriented programming, platform independence, security, robustness, multithreading, and high performance.



Features of Java

- **Object-Oriented Programming (OOP):** Java supports OOP concepts to create modular and reusable code.
- **Platform Independence:** Java programs can run on any operating system with a JVM.
- **Robust and Secure:** Java ensures reliability and security through strong memory management and exception handling.

- **Multithreading and Concurrency:** Java allows concurrent execution of multiple tasks for efficiency.
- **Rich API and Standard Libraries:** Java provides extensive built-in libraries for various programming needs.
- **Frameworks for Enterprise and Web Development:** Java supports frameworks that simplify enterprise and web application development.
- **Open-Source Libraries:** Java has a wide range of libraries to extend functionality and speed up development.
- **Maintainability and Scalability:** Java's structured design allows easy maintenance and growth of applications.

## Understanding the Hello World Program in Java

When we learn any programming language, the first step is writing a simple program to display "Hello World". So, here is a simple Java program that displays "Hello World" on the screen.

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello World!");  
    }  
}
```

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### Output

Hello World!

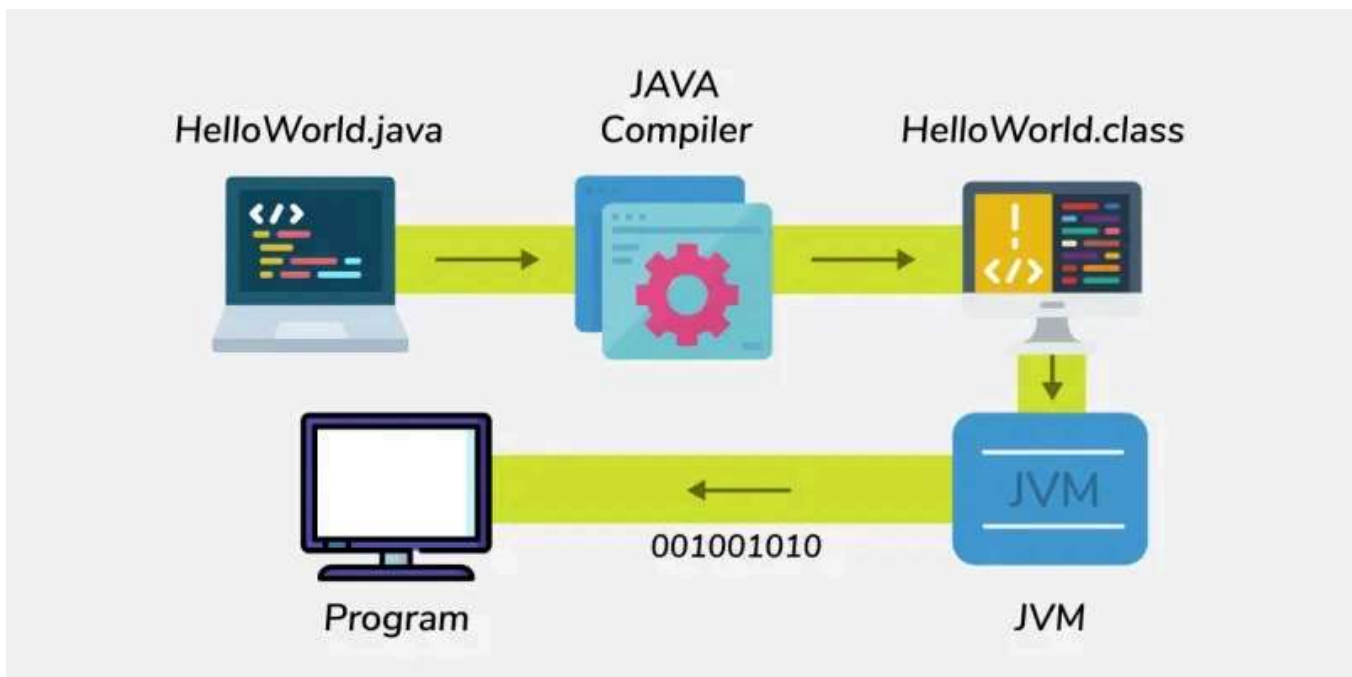
### Explanation:

|                                                      |                                                               |
|------------------------------------------------------|---------------------------------------------------------------|
| <code>// A Java program to print Hello World!</code> | This is a single-line comment                                 |
| <code>public class HelloWorld</code>                 | Defines a class named HelloWorld.                             |
| <code>{</code>                                       | There is an Opening Bracket for Class                         |
| <code>public static void main(String[] args)</code>  | This is the main method, the entry point of any Java program. |
| <code>{</code>                                       | There is an Opening Bracket for Method                        |
| <code>System.out.println("Hello World!");</code>     | Output Statement prints "Hello World!" to the console.        |
| <code>}</code>                                       | There is a Closing Bracket for Method                         |
| <code>}</code>                                       | There is a Closing Bracket for Class                          |

Hello World Program Explanation

## How to run the above code?

- Write code in a file like HelloWorld.java.
- Java Compiler "javac" compiles it into bytecode "HelloWorld.class".
- [JVM](#) (Java Virtual Machine) reads the .class file and interprets the bytecode.
- The JVM initially interprets bytecode and uses JIT (Just-In-Time) compilation to convert frequently executed code into native machine code for better performance.



Java Program Execution Flow

## Comments in Java

Comments are the notes written inside the code to explain what we are doing. The comment lines are not executed while we run the program.

## Types of Comments

**1. Single-line comment:** Used to write a comment in a single line.

```
// This is a comment
```

**2. Multi-line comment:** Used to write comments over multiple lines.

```
/*  
This is a multi-line comment.  
This is useful for explaining larger sections of code.  
*/
```

## Curly Braces and Indentation in Java

In Java, curly braces {} are used to define blocks of code. For example, the body of a class or method is enclosed within curly braces.

Example:

```
public class Geeks{  
    public static void main(String[] args) {  
        {  
            System.out.println("This is inside the block.");  
        }  
        System.out.println("This is outside the block.");  
    }  
}
```

- The code block inside {} runs unconditionally and prints a message.
- Curly braces {} group multiple statements into a single block, even without conditions.

## Famous Applications Built Using Java

- **Android Apps:** Most of the Android mobile apps are built using Java.
- **Netflix:** This uses Java for content delivery and backend services.
- **Amazon:** Java language is used for its backend systems.
- **LinkedIn:** This uses Java for handling high traffic and scalability.
- **Minecraft:** This is one of the world's most popular games that is built in Java.
- **Spotify:** This uses Java in parts of its server-side infrastructure.
- **Uber:** Java is used for backend services like trip management.
- **NASA WorldWind:** This is a virtual globe software built using Java.

## Uses of Java

Java is used for:

- **Mobile App Development:** Android development using Android Studio.
- **Web Development:** Using frameworks like [Spring](#), [Spring Boot](#), [Struts](#), [Hibernate](#)
- **Desktop GUI Applications:** With libraries like [JavaFX](#) and [Swing](#).
- **Enterprise Applications:** Backbone of banking, ERP and large-scale business software.
- **Game Development:** With engines like LibGDX and jMonkeyEngine.
- **Big Data Technologies:** Like [Hadoop](#) and [Apache Kafka](#).
- **Internet of Things (IoT):** Java can run on embedded systems and devices.
- **Cloud-based Applications:** Java is used in services on [AWS](#), [Azure](#) and [Google Cloud](#).
- **Scientific Applications:** Java is used in tools that process large amounts of scientific data.

**Related Article:** [Java Naming Convention](#)

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### Suggested Quiz

 6 Questions

What is used to compile Java source code?

- ☐ A javac
- ☐ B java
- ☐ C jdk

D

jvm

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A-143, 7th Floor, Sovereign Corporate Tower, Sector- 136, Noida, Uttar Pradesh (201305)

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